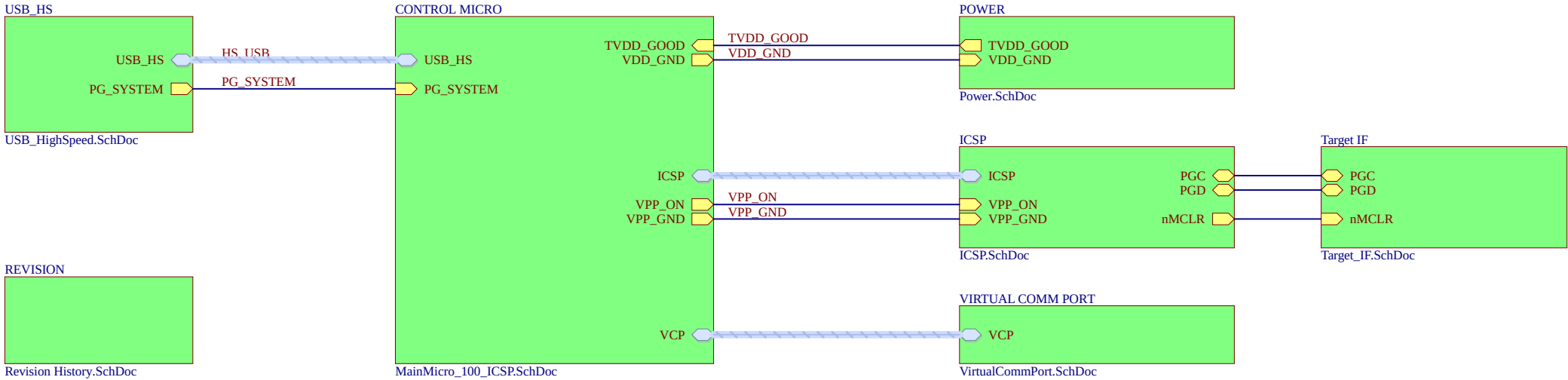


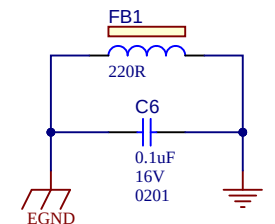
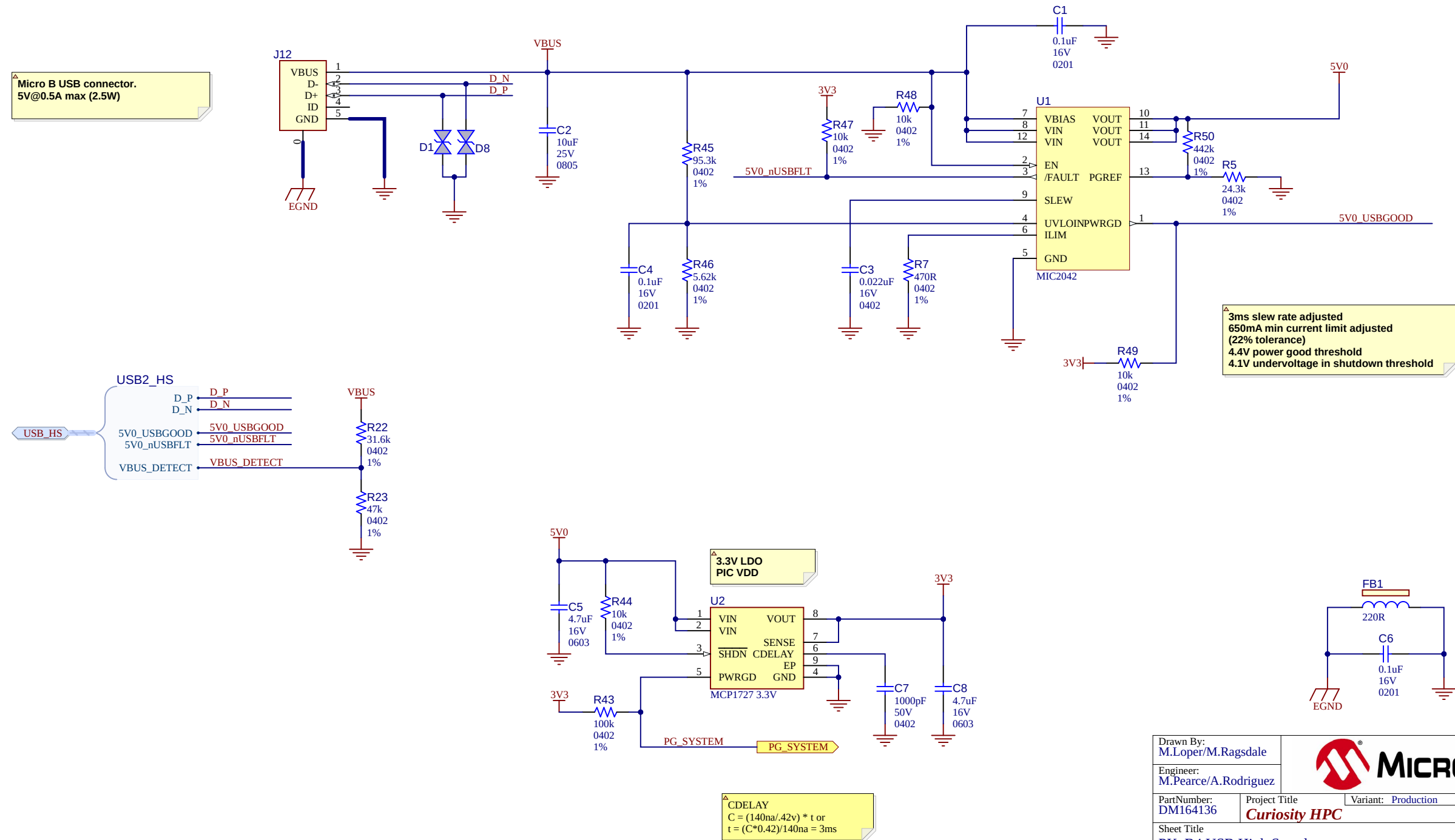
Main System Interconnect

Curiosity HPC

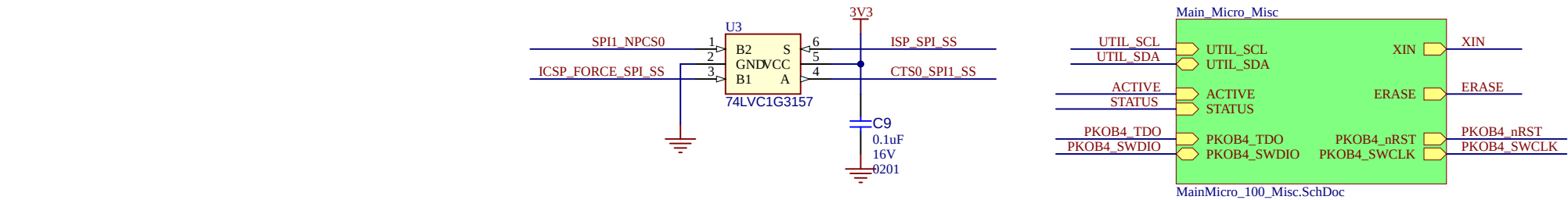
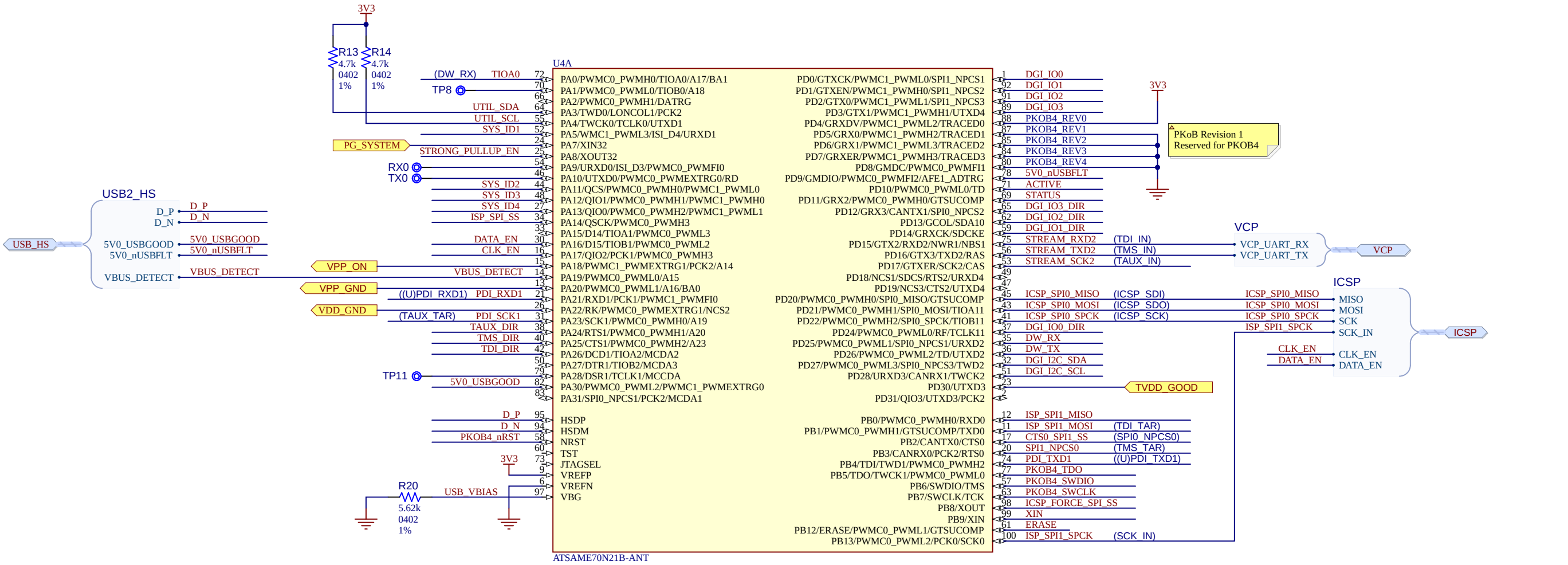


Drawn By: M.Loper/M.Ragsdale		 MICROCHIP	
Engineer: M.Pearce/A.Rodriguez			
PartNumber: DM164136		Project Title Curiosity HPC	Variant: Production
Sheet Title PKoB4 Main Block			
Size B	Sch #:03-10536	Date: 10/29/2020 7:50:15 PM	<i>Designed with</i>  Altium.com
File: 03-10536.SchDoc			

USB High Speed

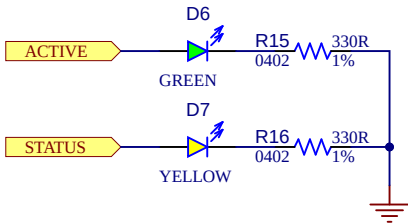
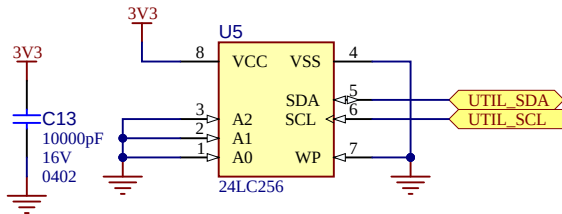
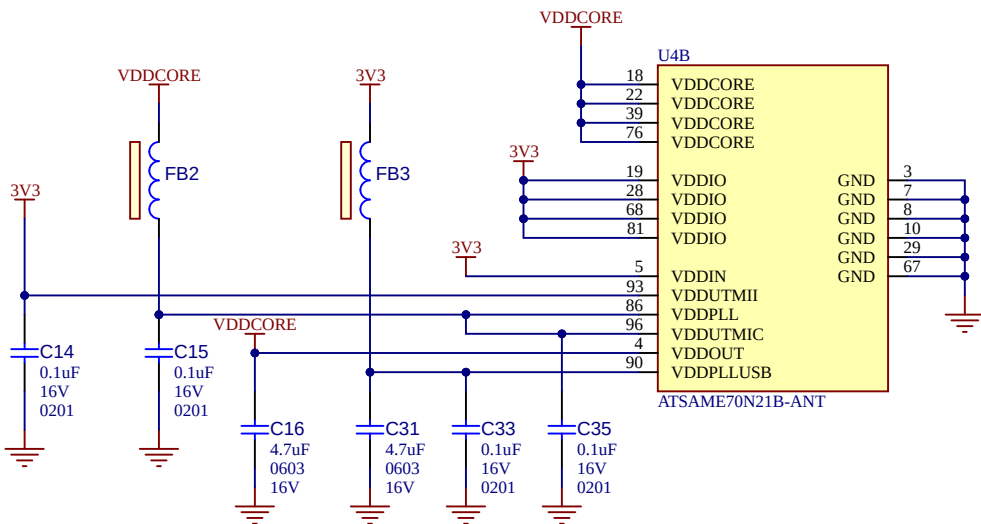
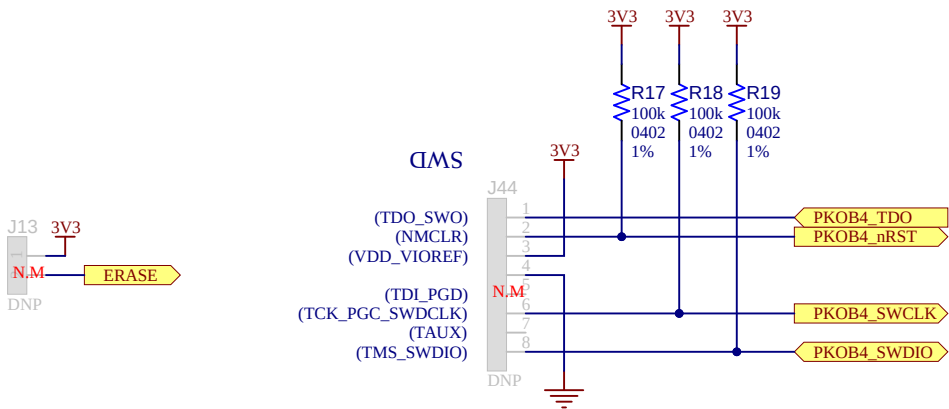


Main Micro

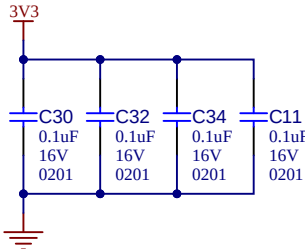


Drawn By: M.Loper/M.Ragsdale			
Engineer: M.Pearce/A.Rodriguez			
PartNumber: DM164136	Project Title Curiosity HPC	Variant: Production	
Sheet Title PKoB4 Main Micro			
Size B	Sch #:03-10536	Date: 10/29/2020 7:50:15 PM	Designed with 
Revision:2		Sheet 3 of 9	Altium.com
File: MainMicro_100_ICSP.SchDoc			

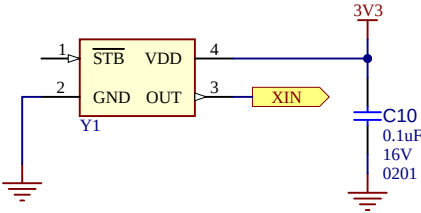
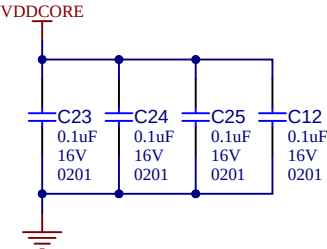
Main Micro Misc.



VDDIO Bypass Caps

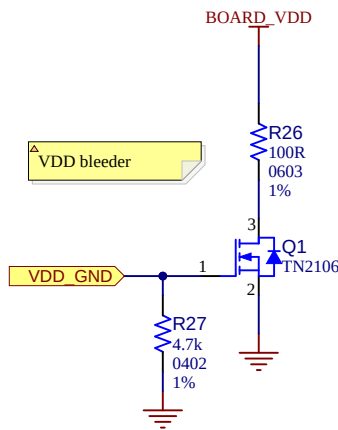
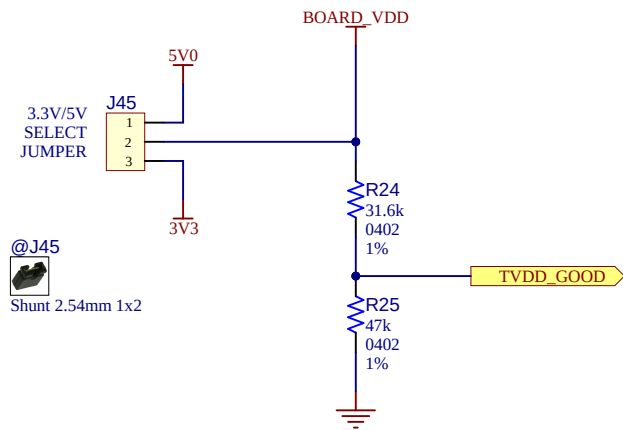


VDDCORE Bypass Caps

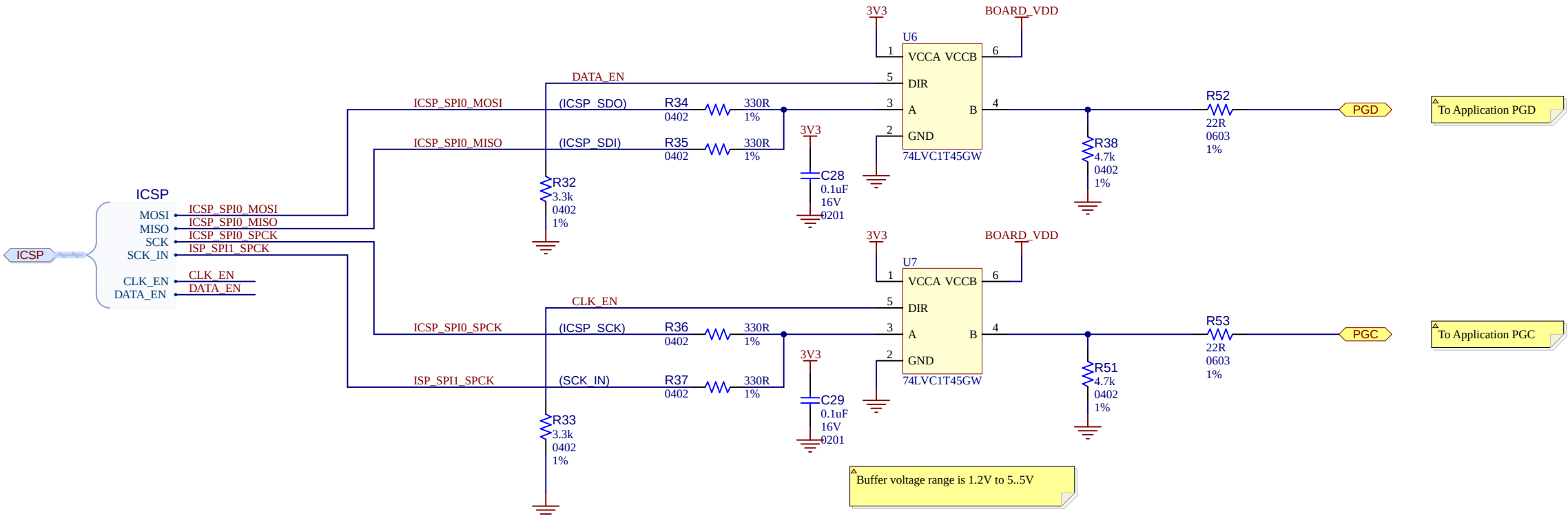


Drawn By: M.Loper/M.Ragsdale			
Engineer: M.Pearce/A.Rodriguez			
PartNumber: DM164136	Project Title Curiosity HPC	Variant: Production	
Sheet Title PKoB4 Main Micro Misc.		Designed with 	
Size B	Sch #:03-10536	Date: 10/29/2020 7:50:15 PM	Sheet 4 of 9
File: MainMicro_100_Misc.SchDoc			

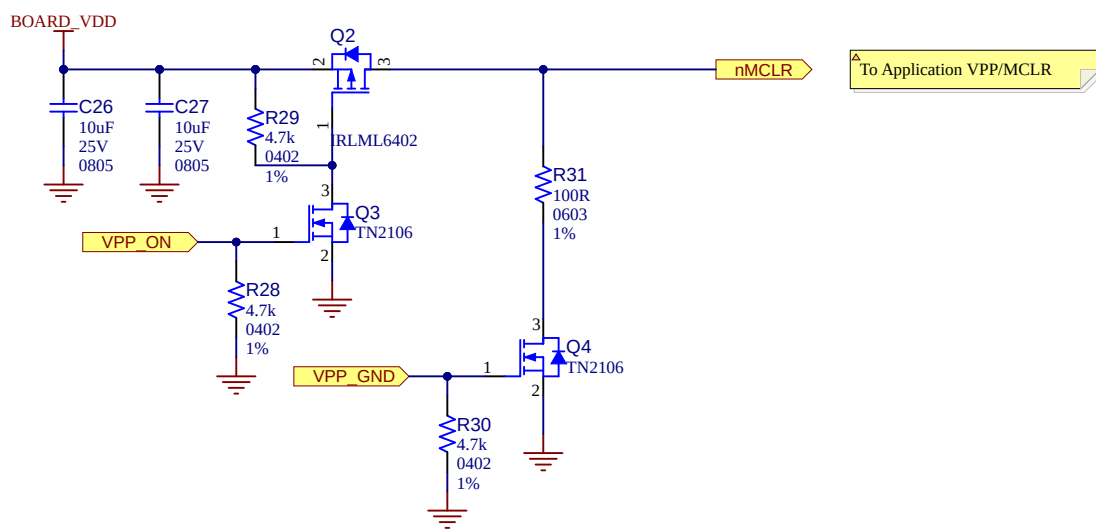
Power



ICSP

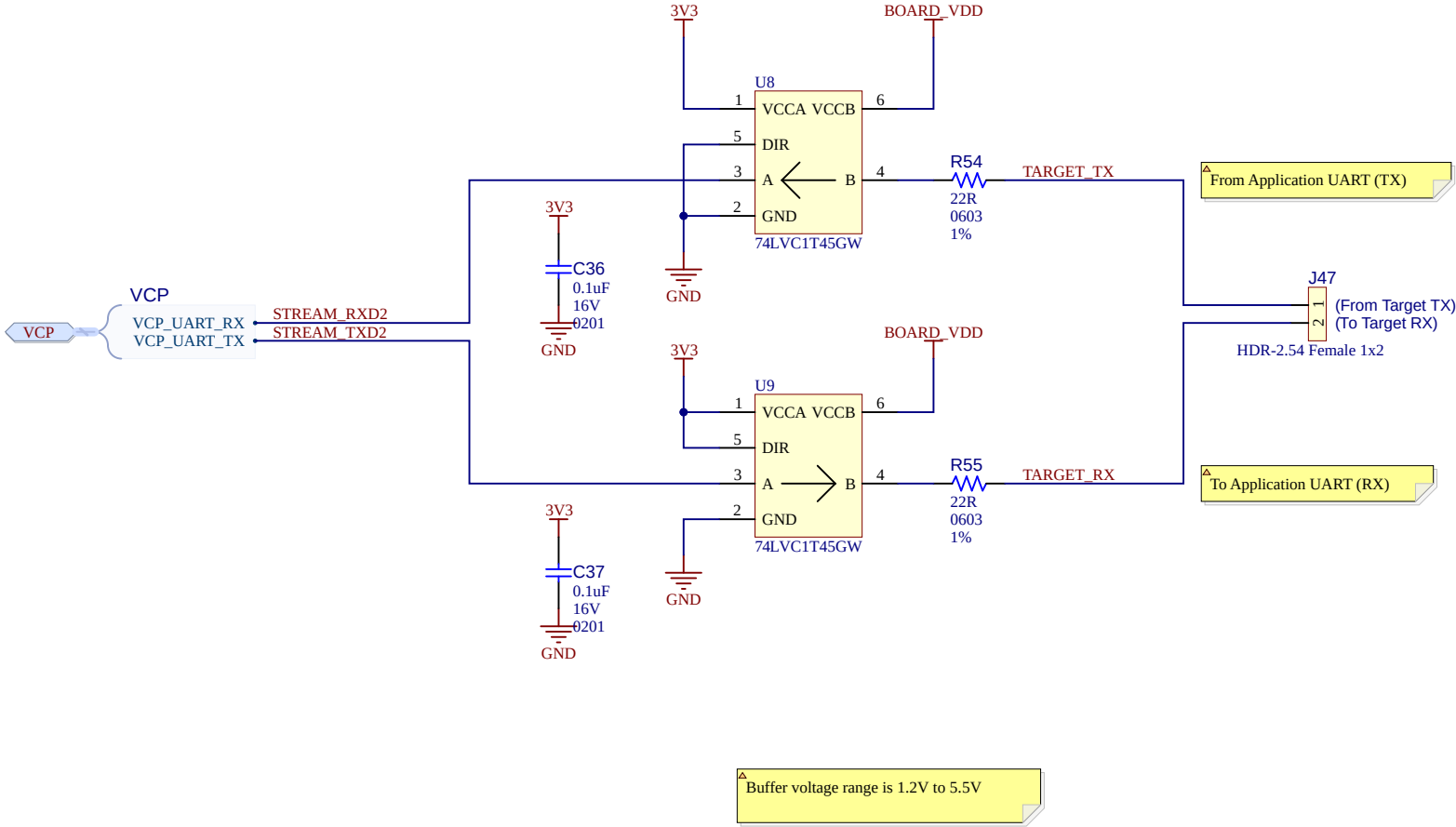


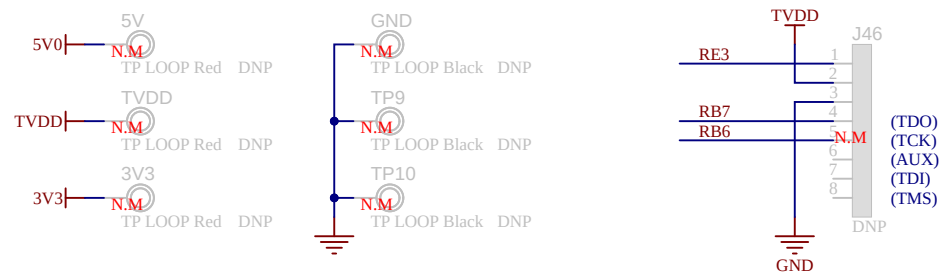
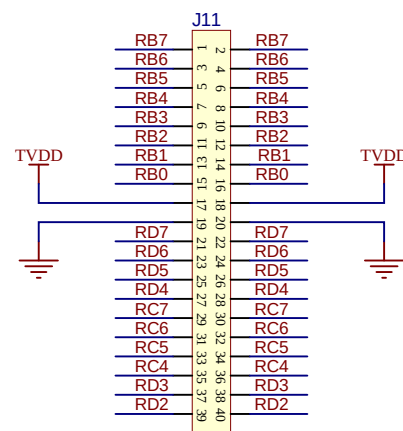
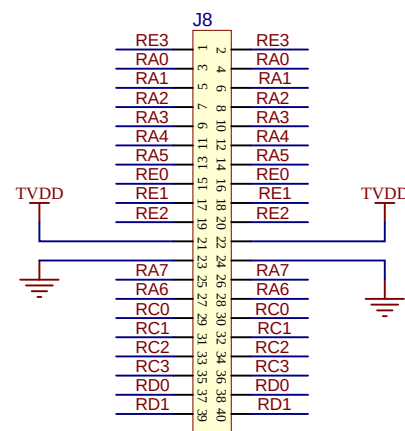
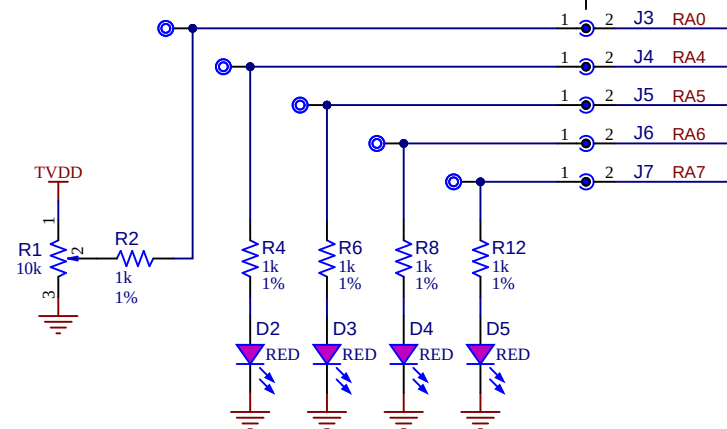
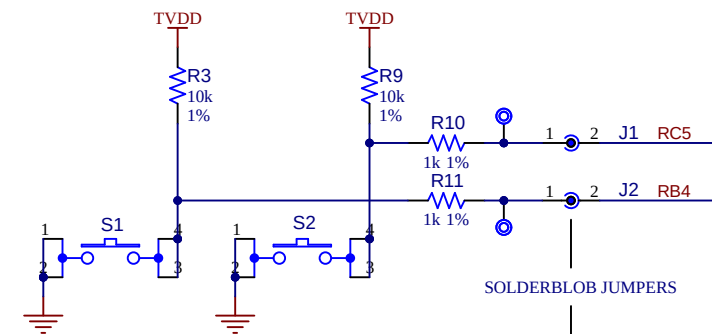
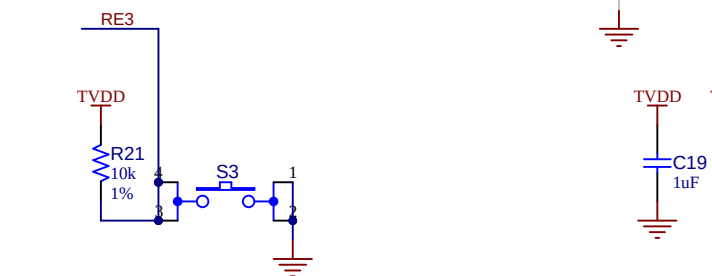
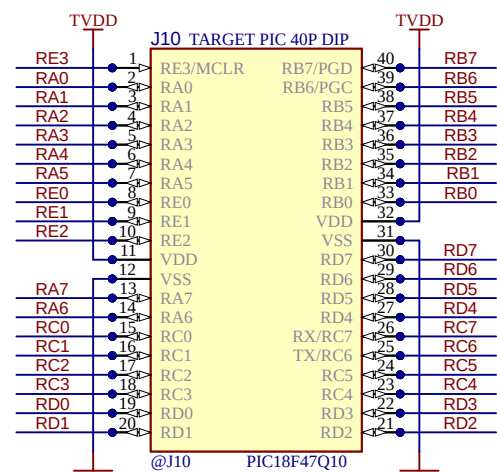
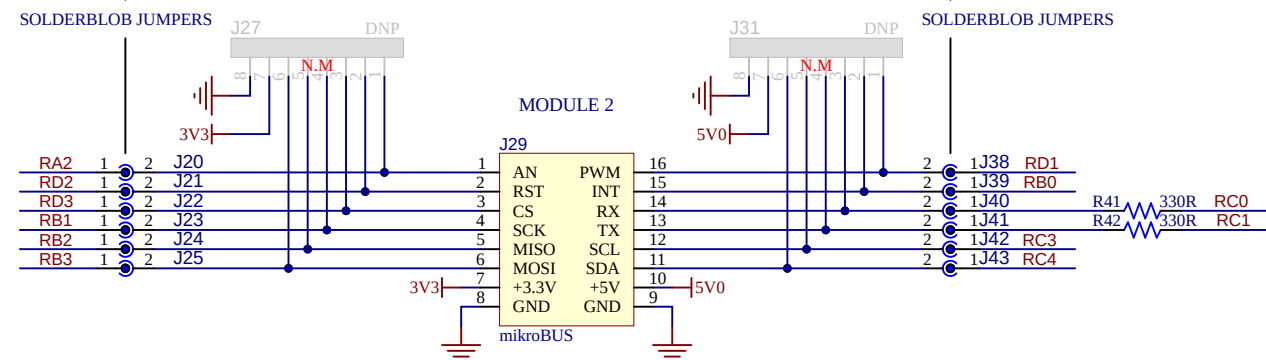
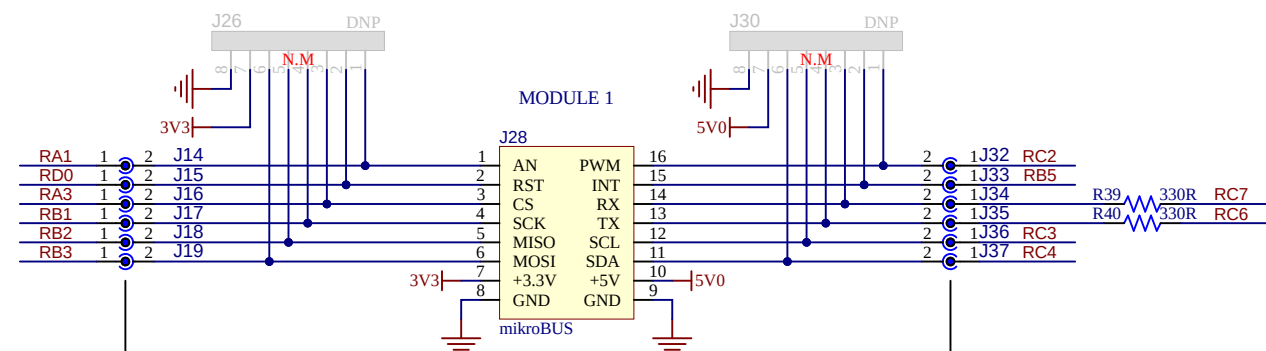
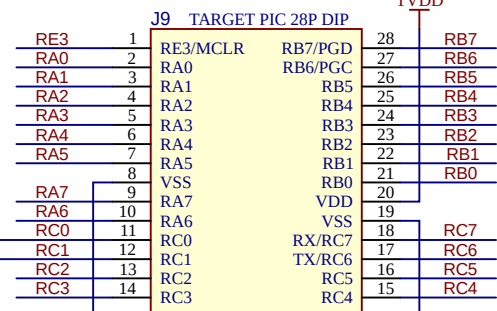
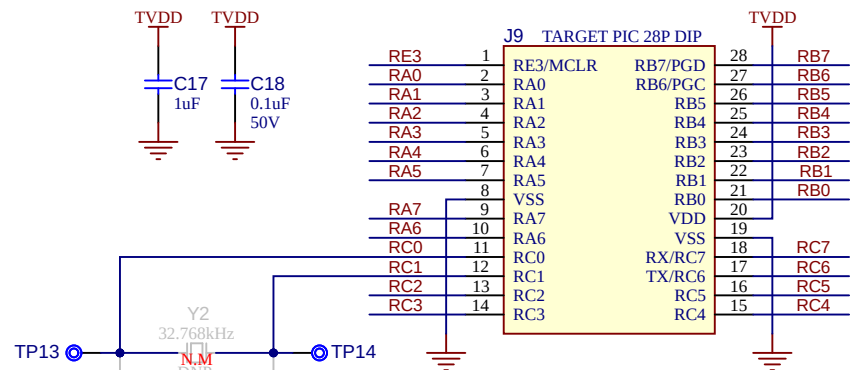
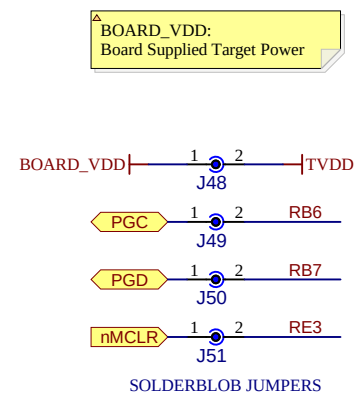
VPP Switch



Drawn By: M.Loper/M.Ragsdale			
Engineer: M.Pearce/A.Rodriguez			
PartNumber: DM164136	Project Title Curiosity HPC	Variant: Production	
Sheet Title ICSP		Designed with 	
Size B	Sch #:03-10536	Date: 10/29/2020 7:50:16 PM	Altium.com
Revision:2		Sheet 6 of 9	
File: ICSP.SchDoc			

Virtual Comm Port





Note: pins are reversed.
Connector is rotated and
mounted on bottom side.

LABEL1

02-
JIT

LABEL Need Help Small

PAD1 PAD2 PAD3 PAD4

Drawn By: M.Loper/M.Ragsdale
Engineer: M.Pearce/A.Rodriguez



MICROCHIP

PartNumber:	Project Title	Variant: Production
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Sheet Title	mikroBUS and Target PIC Interface
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Size B	Sch #:03-10536	Date: 10/29/2020 7:50:16 PM
	Revision:2	Sheet 8 of 9

File: Target IF.SchDoc

Designed with
Altium
Altium.com

Revision History

Rev. A: **2016-02-17**

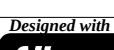
Initial Design

Rev. 1: **2016-03-31**

Rev. 1.1: **2016-06-23**

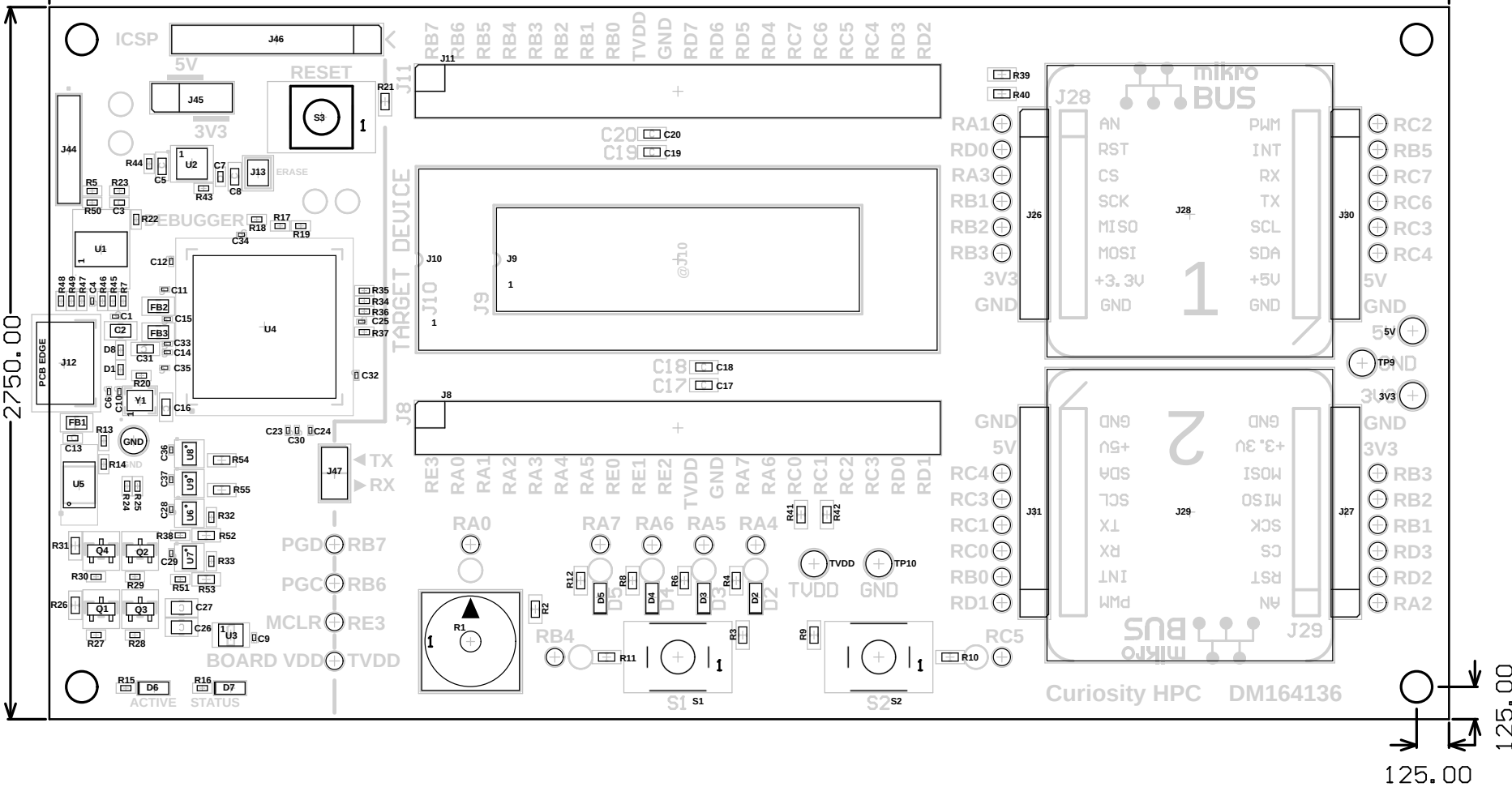
Rev. 2: **2019-10-25**

Updated to use PkoB 4 debugger design.
All components updated to MCL standards.

Drawn By: M.Loper/M.Ragsdale		 MICROCHIP		
Engineer: M.Pearce/A.Rodriguez				
PartNumber: DM164136	Project Title Curiosity HPC	Variant: Production		
Sheet Title Revision History		 Altium Altium.com		
Size B	Sch #03-10536			Date: 10/29/2020 7:50:17 PM
	Revision:2			Sheet 9 of 9
File: Revision History.SchDoc				

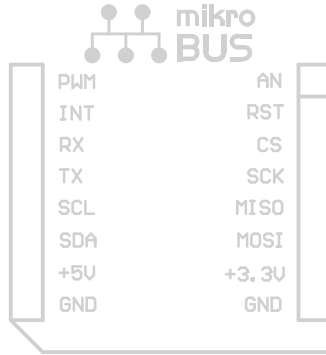
Board Outline

Board Outline

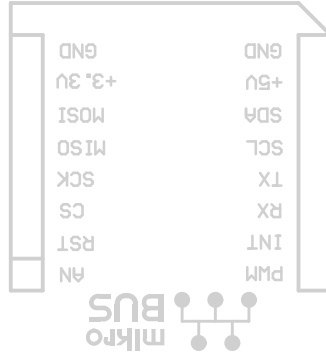
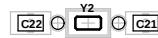


04-10536 Rev

Rubber
Pad



Rubber
Pad



LABEL1



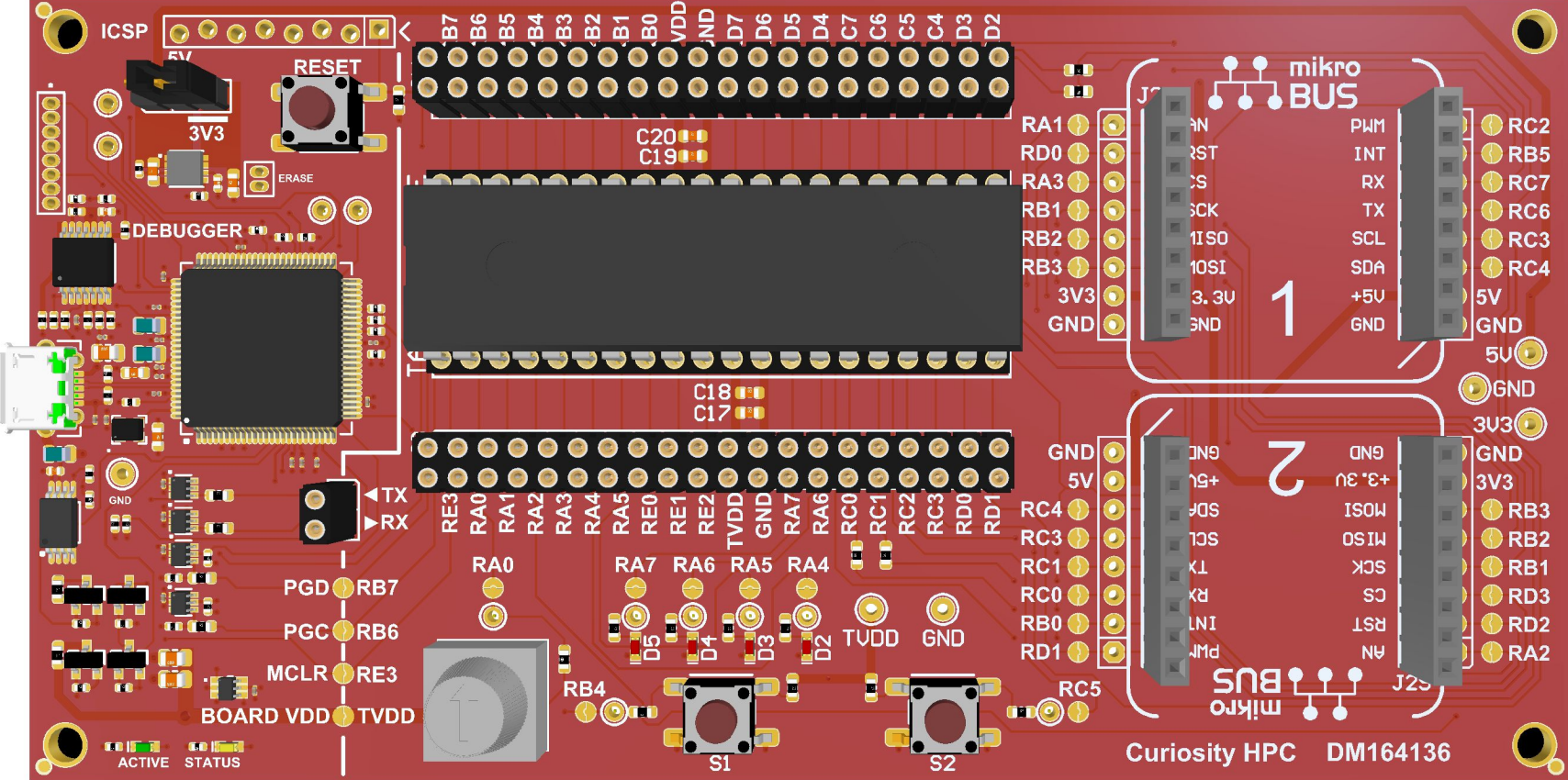
www.microchip.com/curiosityhpc



Rubber
Pad

Rubber
Pad

Bottom Side



04-10536 Rev

04-10536 Rev

mikro
BUS

PWM	AN
INT	RST
RX	CS
TX	SCK
SCL	MISO
SDA	MOSI
+5V	+3.3V
GND	GND

GND	GND
+3.3V	+5V
MOSI	SDA
MISO	SCL
SCK	TX
CS	RX
RST	INT
AN	PWM

mikro
BUS

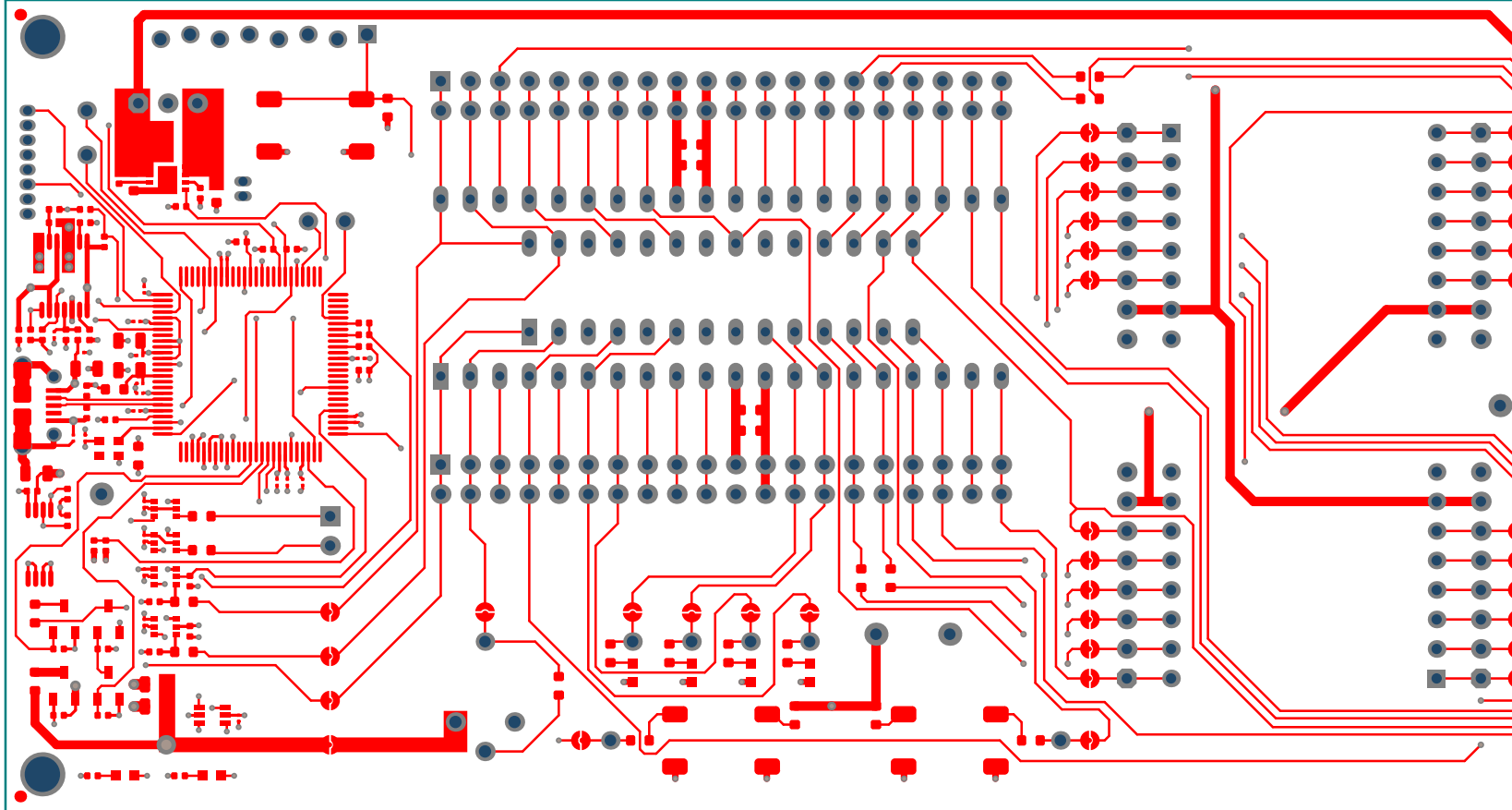
LABEL1

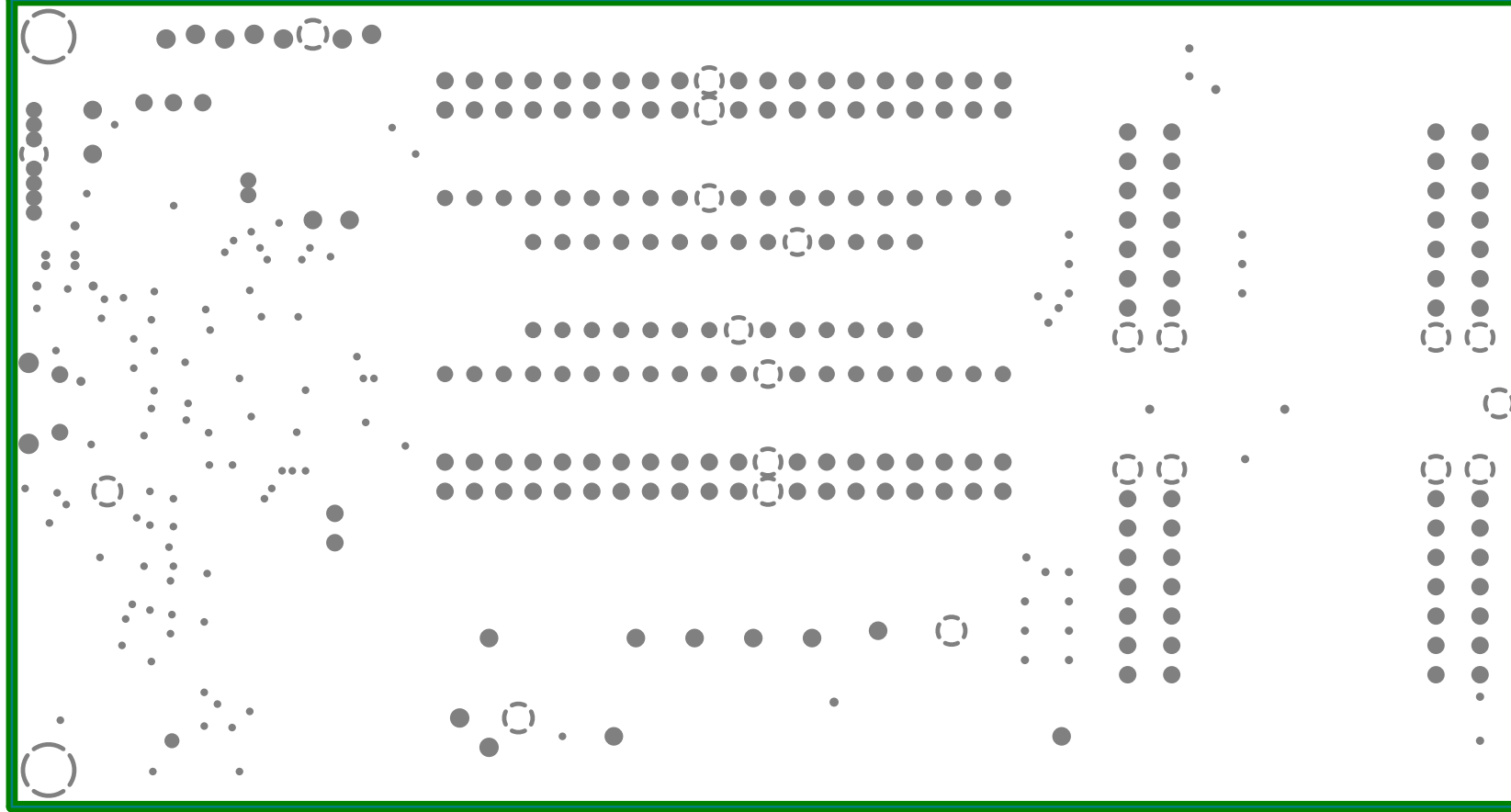


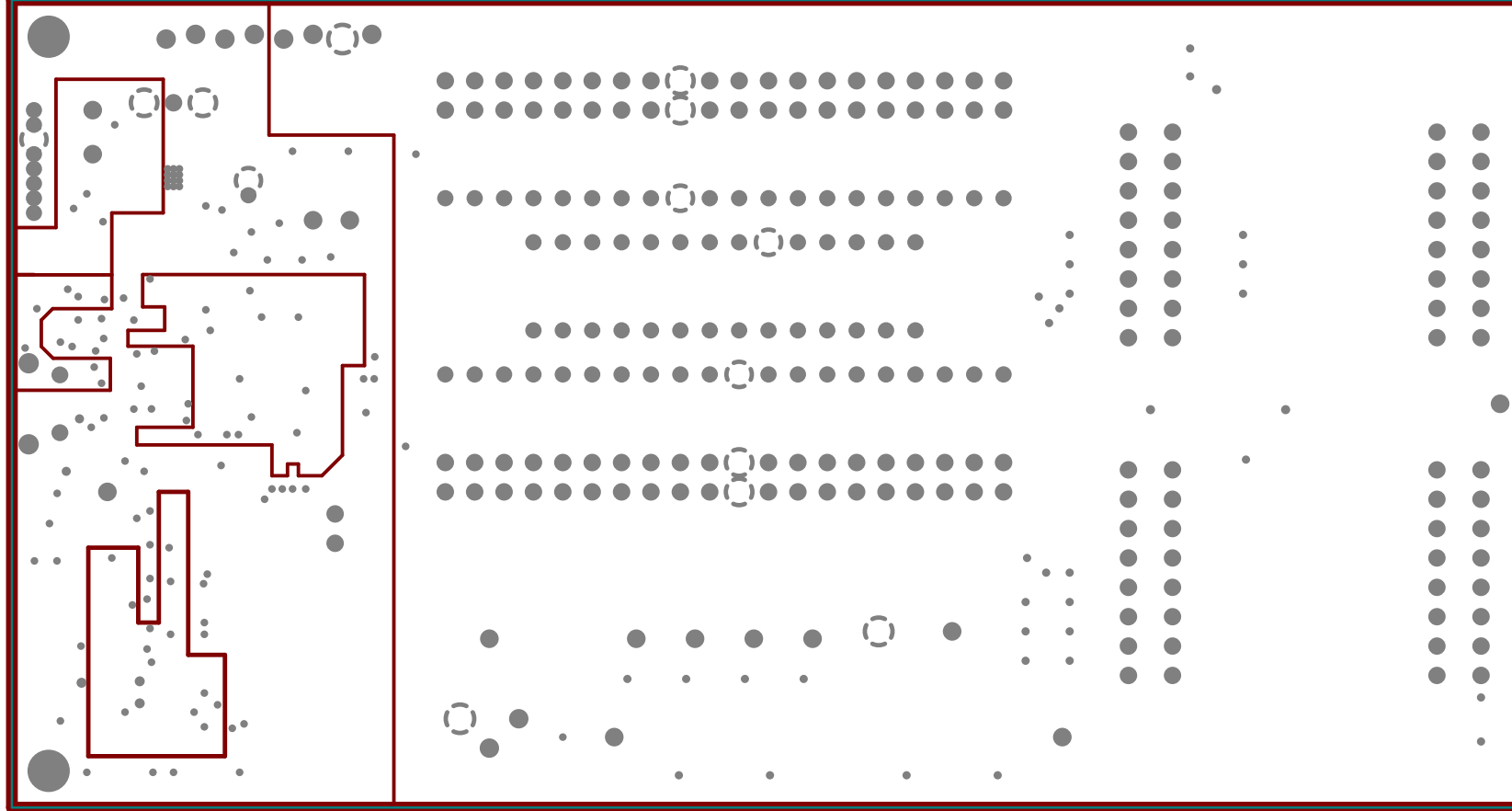
www.microchip.com/curiosityhpc

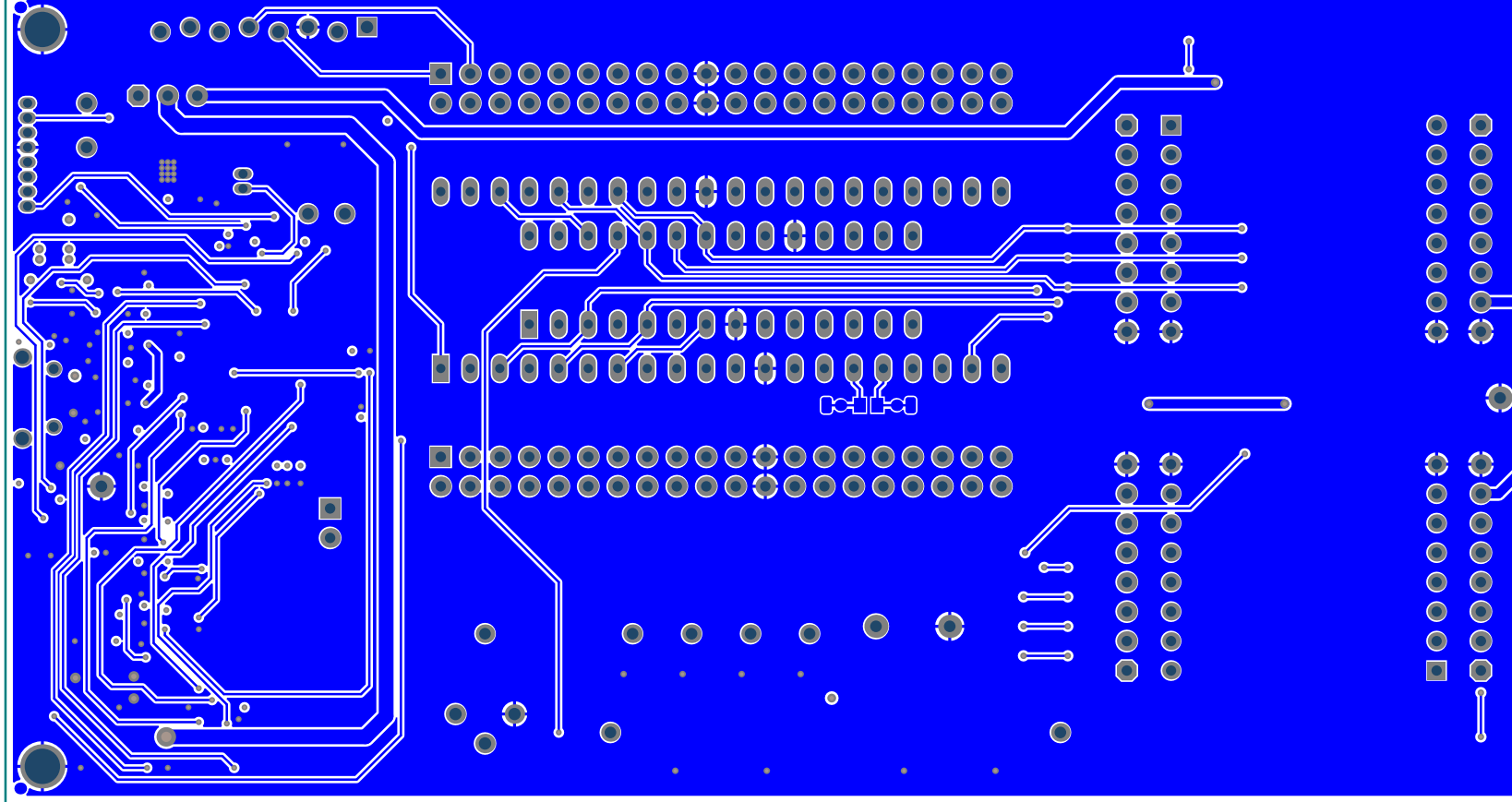

MICROCHIP











Component list

Bill of Materials Fitted for Variant [Production] of Project [High Pin Count Curiosity.PrjPcb] (No PCB Document Selected)

Source Data From:	High Pin Count Curiosity.PrjPcb
Project:	High Pin Count Curiosity.PrjPcb
Variant:	Production



Report Date: 10/29/2020 7:50 PM
Print Date: 10/29/2020 7:50:27 PM

Fitted	Designator	Quantity	Value	Manufacturer 1	Manufacturer Part Number 1	Description
Fitted	@J10	1	PIC18F47Q10	Microchip Technology	PIC18F47Q10-VP	Microcontroller
Fitted	@J45	1	Shunt 2.54mm 1x2	Würth Electronics Inc.	60900213421	
Fitted	C1, C4, C6, C9, C10, C11, C12, C14, C15, C23, C24, C25, C28, C29, C30, C32, C33, C34, C35, C36, C37	21	0.1uF	Murata Electronics North America	GRM033R61C104KE84D	Capacitor
Fitted	C2, C26, C27	3	10uF	TDK Corporation	C2012X5R1E106K125AB	Capacitor
Fitted	C3	1	0.022uF	Samsung Electro-Mechanics America, Inc	CL05B223KO5NNNC	Capacitor
Fitted	C5, C8, C16, C31	4	4.7uF	TDK Corporation	C1608X5R1C475K080AC	Capacitor
Fitted	C7	1	1000pF	Murata	GRM155R71H102KA01D	Capacitor
Fitted	C13	1	10000pF	KEMET	C0402C103K4RACTU	Capacitor
Fitted	C17, C19	2	1uF	Taiyo Yuden	LMK107B7105KA-T	Capacitor
Fitted	C18, C20	2	0.1uF	TDK	C1608X7R1H104M	Capacitor
Fitted	D1, D8	2	PGB101	Littlefuse	PGB1010402KR	
Fitted	D2, D3, D4, D5	4	RED	Kingbright	APT1608EC	LED
Fitted	D6	1	GREEN	Lite-On Inc	LTST-C191KGKT	LED
Fitted	D7	1	YELLOW	Lite-On	LTST-C190YKT	LED
Fitted	FB1	1	220R	Murata Electronics North America	BLM21PG221SN1D	FERRITE 180R 1.5A SMD 0603
Fitted	FB2, FB3	2	600R	TDK Corporation	MPZ2012S601AT000	FERRITE 180R 1.5A SMD 0603
Fitted	J8, J11	2	HDR-2.54 Female 2x20	Mil-Max	803-43-040-10-001000	
Fitted	J9	1	110-91-328-41-001	Mil-Max	110-43-328-41-001000	Connector
Fitted	J10	1	110-87-640-41-001151	Preci-Dip	110-87-640-41-001151	Connector
Fitted	J12	1	USB2.0 MICRO-B FEMALE	FCI	10118194-0001LF	
Fitted	J28, J29	2	mkroBUS	Sullins Connector Solutions	PPTC081LFBN-RC	
Fitted	J45	1	HDR-2.54 Male 1x3	FCI	68000-103HLF	
Fitted	J47	1	HDR-2.54 Female 1x2	Mil-Max	803-43-002-10-001000	
Fitted	LABEL 1	1	LABEL Need Help Small			
Fitted	PAD1, PAD2, PAD3, PAD4	4	Rubber Pad Cyl D7.9H5.3	3M	SJ61A11	
Fitted	Q1, Q3, Q4	3	TN2106	Microchip Technology	TN2106K1-G	Transistor MOS
Fitted	Q2	1	IRLML6402	International Rectifier	IRLML6402TRPBF	Transistor MOS
Fitted	R1	1	10k	TT Electronics/BI TT Electronics/BI	72PTR10KLF	Resistor Var
Fitted	R2, R4, R6, R8, R10, R11, R12	7	1k	Panasonic	ERJ-3BK1001V	Resistor
Fitted	R3, R9, R21	3	10k	Panasonic	ERJ-3BK1002V	Resistor
Fitted	R5	1	24.3k	Samsung	RC1005F2432CS	Resistor
Fitted	R7	1	470R	Yageo	RC0402FR-07470RL	Resistor
Fitted	R13, R14, R27, R28, R29, R30, R38, R51	8	4.7k	Yageo	RC0402FR-074K7L	Resistor
Fitted	R15, R16, R34, R35, R36, R37	6	330R	Yageo	RC0402FR-07330RL	Resistor
Fitted	R17, R18, R19, R43	4	100k	Panasonic	ERJ-2RK1003X	Resistor
Fitted	R20, R46	2	5.62k	Vishay Dale	CRCW04025K62FKED	Resistor
Fitted	R22, R24	2	31.6k	Panasonic	ERJ-2RK3162X	Resistor
Fitted	R23, R25	2	47k	Panasonic - ECG	ERJ-2RK4702X	Resistor
Fitted	R26, R31	2	100R	ROHM	MCR03EZPFX1000	Resistor
Fitted	R32, R33	2	3.3k	Panasonic - ECG	ERJ-2RK3301X	Resistor
Fitted	R39, R40, R41, R42	4	330R	Panasonic	ERJ-3BK3300V	Resistor
Fitted	R44, R47, R48, R49	4	10k	Panasonic	ERJ-2RK1002X	Resistor
Fitted	R45	1	95.3k	Rohm Semiconductor	MCR01MRTF9532	Resistor
Fitted	R50	1	442k	Samsung Electro-Mechanics America, Inc	RC1005F4423CS	Resistor
Fitted	R52, R53, R54, R55	4	22R	Yageo	RC0603FR-0722RL	Resistor
Fitted	S1, S2, S3	3	PTS645SM43SMTR92 LFS	C&K Components	PTS645SM43SMTR92 LFS	Switch
Fitted	U1	1	MIC2042	Microchip Technology	MIC2042-1YTS	IC SMT, MIC2042, DISTRIBUTION SW SGL 14-TSSOP
Fitted	U2	1	MCP1727 3.3V	Microchip Technology	MCP1727-3302EMF	LDO
Fitted	U3	1	74LVC1G3157	Texas Instruments	SN74LVC1G3157DKR	
Fitted	U4	1	ATSAME70N21B-ANT	Microchip Technology	ATSAME70N21B-ANT	
Fitted	U5	1	24LC256	Microchip Technology	24LC256T-E/ST	EEPROM I2C
Fitted	U6, U7, U8, U9	4	74LVC1T45GW	NXP USA Inc.	74LVC1T45GW.125	
Fitted	Y1	1	12.00 MHz	Microchip Technology Inc	DSC6011J11A-012.0000	
Not Fitted	3V3, 5V, TVDD	0	TP LOOP Red	Keystone	5000	Test Point
Not Fitted	C21, C22	0	22pF	Johanson Dielectrics	500R15N220JV4T	Capacitor
Not Fitted	GND, TP9, TP10	0	TP LOOP Black	Keystone	5001	Test Point
Not Fitted	J13	0	HDR-1.27 Male 1x2	Digikey	952-3598-ND	
Not Fitted	J26, J27, J30, J31	0	HDR-2.54 Male 1x8	FCI	68001-108HLF	CON HDR-2.54 Female 1x8 Gold TH
Not Fitted	J44	0	HDR-1.27 Female 1x8	Greenconn	FSEA120-0802A002B1AB	CON HDR-2.54 Female 1x8 Gold TH
Not Fitted	J46	0	HDR-2.54 Male 1X8 STAGGERED			CON HDR-2.54 Female 1x8 Gold TH
Not Fitted	Y2	0	32.768kHz	NDK	NX2012SA-32.768K-STD-MUB-1	Crystal

	134	
Approved	Notes	